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Scattering mechanisms and electronic transport in conductors

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A. Units and notations

$k_B = 1, \hbar = 1$ (unless specified otherwise).

Energies are measured in Kelvin or in inverse seconds, depending on the situation. Useful conversion relations:

$$\begin{aligned} 1 \text{ ev} &\approx 10^4 \text{ K} \\ 1 \text{ K} &\approx 10^{11} \text{ s}^{-1}. \end{aligned}$$

Units of the wavevector and momentum are the same.

$\simeq$ stands for “equal in order of magnitude”. For example, $1 \simeq 2, 10 \simeq 12$.

$\approx$ stands for “close to”. $a \approx b$ means that $|a - b| \ll \min\{a, b\}$. For example, $1.1 \approx 1$

$\propto$ stands for “proportional to” or “scales as”. For example, the specific heat of a Fermi gas $C(T) \propto T$.

Useful relations

In metals,

$$E_F \simeq e^2/a_0 \simeq M\omega_D^2 \simeq 1 \text{ ev} \approx 10^{15} \text{ s}^{-1},$$

where a_0 is the lattice spacing and ω_D is the Debye frequency. Also,

$$a_0 \simeq 1/me^2 \simeq 10^{-8} \text{ cm}.$$

I. BOLTZMANN EQUATION

Boltzmann equation describes the evolution of the *distribution function* in space, momentum, and time.

Definition 1 *The distribution function, $f(\mathbf{r}, \mathbf{p}, t)$, is such that a quantity $f(\mathbf{r}, \mathbf{p}, t) d^3r d^3p$ gives the number of particles located in volume d^3r near point $\mathbf{r}$ whose momenta occupy volume d^3p near a point in the momentum space $\mathbf{p}$ at time t .*

Obviously,

$$\int \frac{d^3p}{(2\pi)^3} f(\mathbf{r}, \mathbf{p}, t) = n(\mathbf{r}, t),$$

where $n(\mathbf{r}, t)$ is the number density and

$$\int d^3r f(\mathbf{r}, \mathbf{p}, t) = n_p(t),$$

where $n_p(t)$ is the (time-dependent) occupation number. Normalization

$$\int d^3r \int \frac{d^3p}{(2\pi)^3} f(\mathbf{r}, \mathbf{p}, t) = N(t),$$

total number of particles in the system. If this number is fixed, the integral on the left-hand-side is time-independent, although the distribution function itself may depend on time. Description in terms of $f(\mathbf{r}, \mathbf{p}, t)$ works for classical systems and for systems where quantum-mechanical effects are small (semi-classical approximation). Obviously, $f(\mathbf{r}, \mathbf{p}, t)$ *cannot* describe a fully quantum-mechanical system, because in such a system one cannot specify the value of the momentum and coordinate simultaneously. In the absence of scattering, $f(\mathbf{r}, \mathbf{p}, t)$ satisfies the *Liouville equation* (see, e.g., Landau & Lifshits, *Statistical Physics*, Ch. 3)

$$\frac{d}{dt} f(\mathbf{r}, \mathbf{p}, t) = 0$$

or, opening the time derivative

$$\begin{aligned} \frac{\partial f}{\partial t} + \frac{d\mathbf{r}}{dt} \cdot \vec{\nabla}_{\mathbf{r}} f + \frac{d\mathbf{p}}{dt} \cdot \nabla_{\mathbf{p}} f &= 0 \\ \frac{\partial f}{\partial t} + \mathbf{v} \cdot \vec{\nabla}_{\mathbf{r}} f + \vec{F} \cdot \nabla_{\mathbf{p}} f &= 0, \end{aligned}$$

where $\vec{F}$ is the (classical) force applied to the system. In the presence of scattering, the RHS of the equation is different from zero because particles change their momentum states due to scattering. These processes are accounted for in the *Boltzmann equation*

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \vec{\nabla}_{\mathbf{r}} f + \vec{F} \cdot \nabla_{\mathbf{p}} f = I_{\text{coll}}[f],$$

where $I_{\text{coll}}[f]$ is the *collision integral*. This term takes into account processes which change a number of particles with given momentum. In a most general form,

$$I_{\text{coll}}[f] = \sum_{\mathbf{p}'} \hat{W}_{\mathbf{p}' \rightarrow \mathbf{p}} - \sum_{\mathbf{p}'} \hat{W}_{\mathbf{p} \rightarrow \mathbf{p}'},$$

where $\hat{W}_{\mathbf{k} \rightarrow \mathbf{k}'}$ is the probability of a corresponding scattering process. Notice that $\hat{W}_{\mathbf{k} \rightarrow \mathbf{k}'}$ involves not only microscopic probabilities but also the distribution function itself. In general, a Boltzmann equation is a *non-linear, integro-differential equation* for f . In case of practical cases on interest, the situation is simplified if one considers a small deviation of equilibrium.

Claim 2 *An equilibrium distribution function, $f_0(\mathbf{p})$, nullifies the collision integral*

$$I_{\text{coll}}[f_0] = 0.$$

Indeed, in thermodynamic equilibrium, the LHS of the Boltzmann equation is equal to zero.

Linearizing the distribution function near its equilibrium value,

$$f = f_0 + \delta f,$$

where $\delta f \ll f_0$, one can solve the equation by successive iterations.

A. Example: conductivity in the presence of impurity scattering.

For scattering of electrons at impurities,

$$I_{\text{coll}}[f] = \sum_{\mathbf{p}'} w_{\mathbf{p}' \rightarrow \mathbf{p}} f_{\mathbf{p}'} (1 - f_{\mathbf{p}}) - f_{\mathbf{p}} (1 - f_{\mathbf{p}'}) w_{\mathbf{p} \rightarrow \mathbf{p}'}.$$

Under quite general conditions, i.e., when both the scattering center and the crystal possess inversion symmetry (see V. F. Gantmakher and Y. B. Levinson, *Scattering of carriers in metals and semiconductors*, Elsevier, 1987) the probability is invariant with respect to an exchange of initial and final states:

$$w_{\mathbf{p}' \rightarrow \mathbf{p}} = w_{\mathbf{p} \rightarrow \mathbf{p}'}.$$

Then

$$\begin{aligned} I_{\text{coll}}[f] &= \sum_{\mathbf{p}'} w_{\mathbf{p} \rightarrow \mathbf{p}'} [f_{\mathbf{p}'} (1 - f_{\mathbf{p}}) - f_{\mathbf{p}} (1 - f_{\mathbf{p}'})] \\ &= \sum_{\mathbf{p}'} w_{\mathbf{p} \rightarrow \mathbf{p}'} (f_{\mathbf{p}'} - f_{\mathbf{p}}) \end{aligned}$$

Notice that the form of the collision integral in this case does not depend on statistics: had we started with the Maxwell-Boltzmann statistics (as a special case of the Fermi-Dirac one) or with the Bose-Einstein one, we would have obtained the same form of I_{coll} .

II. THE SIMPLEST FORM OF THE COLLISION INTEGRAL: τ - APPROXIMATION.

The most popular form of the collision integral is the τ - approximation. It works for i) elastic and ii isotropic scattering, when $w_{\mathbf{p} \rightarrow \mathbf{p}'}$ can be chosen as

$$w_{\mathbf{p} \rightarrow \mathbf{p}'} = \frac{1}{L^3 \nu(\varepsilon_p)} \delta(\varepsilon_p - \varepsilon_{p'}) \frac{1}{\tau},$$

where τ is the *phenomenologically* introduced scattering time. Then,

$$\begin{aligned} I_{\text{coll}}[f] &= \sum_{\mathbf{p}'} w_{\mathbf{p} \rightarrow \mathbf{p}'} (f_{\mathbf{p}'} - f_{\mathbf{p}}) = L^3 \int \frac{d^3 p'}{(2\pi)^3} \frac{1}{L^3 \nu(\varepsilon_{p'})} \delta(\varepsilon_p - \varepsilon_{p'}) \frac{1}{\tau(\varepsilon_p)} (f_{\mathbf{p}'} - f_{\mathbf{p}}) \\ &= \int d\varepsilon_{p'} \nu(\varepsilon_{p'}) \delta(\varepsilon_p - \varepsilon_{p'}) \frac{1}{\nu(\varepsilon_{p'}) \tau} \int \frac{d\Omega'}{4\pi} (f_{\mathbf{p}'} - f_{\mathbf{p}}) = \frac{1}{\tau(\varepsilon_p)} \int \frac{d\Omega'}{4\pi} (f_{\mathbf{p}'} - f_{\mathbf{p}}) \\ &= \frac{1}{\tau(\varepsilon_p)} (\bar{f} - f_{\mathbf{p}}), \end{aligned}$$

where $\bar{f}$ is the distribution function averaged over directions of $\mathbf{p}'$.

Assume that we apply a constant-in-time electric field to the system, $\mathbf{E}$, and consider a steady-state, uniform case, i.e., a time-independent situation, when f is a function of $\mathbf{p}$ only. Physically, it means that we apply a constant electric field to the system, wait until the transient currents subside and measure a dc current as a response to the field. As the electric field is homogeneous through the system, so is f . The LHS of the Boltzmann equation reduces to

$$\frac{df}{dt} = \vec{F} \cdot \vec{\nabla}_p f = -e\mathbf{E} \cdot \vec{\nabla}_p f.$$

Representing f as

$$f = f_0 + \mathbf{E} \cdot \mathbf{v} \chi,$$

so that $\bar{f} = f_0$. We have written the correction to f_0 in a way that manifests its proportionality to the (weak) electric field. The form $\mathbf{E} \cdot \mathbf{v}$ is the only scalar one can construct out of two vectors; $\mathbf{v}$ and $\mathbf{E}$. Because the angular dependence of the correction is already specified, χ depends only on ε . To the first order in $\mathbf{E}$, we neglect χ in the LHS and take into account that f_0 depends only on the magnitude but not direction of $\mathbf{p}$. Specifying the magnitude of $\mathbf{p}$ is the same as specifying the electrons' energy $\varepsilon = p^2/2m$. Therefore,

$$\begin{aligned} -e\mathbf{E} \cdot \vec{\nabla}_p f &\approx -e\mathbf{E} \cdot \vec{\nabla}_p f_0 = -e\mathbf{E} \cdot \vec{\nabla}_p f_0(\varepsilon) \\ &= -e\mathbf{E} \cdot \vec{\nabla}_p \varepsilon \frac{\partial f_0}{\partial \varepsilon} = -e\mathbf{E} \cdot \mathbf{v} \frac{\partial f_0}{\partial \varepsilon}. \end{aligned}$$

The Boltzmann equation now reduces to an algebraic one

$$\begin{aligned} -e\mathbf{E} \cdot \mathbf{v} \frac{\partial f_0}{\partial \varepsilon} &= -\mathbf{E} \cdot \mathbf{v} \frac{\chi}{\tau} \\ &\rightarrow \chi = e\tau \frac{\partial f_0}{\partial \varepsilon} \end{aligned}$$

Electric current

$$\vec{j} = -2e \int \frac{d^3p}{(2\pi)^3} \mathbf{v} f = 2e^2 \int \frac{d^3p}{(2\pi)^3} \tau \mathbf{v} (\mathbf{E} \cdot \mathbf{v}) \left(-\frac{\partial f_0}{\partial \varepsilon} \right),$$

where 2 comes from summing over two spin projections. The equilibrium function drops out from the result for the current. Suppose that $\mathbf{E}$ is along the z -axis. Then

$$j_z = 2e^2 \int \frac{d^3p}{(2\pi)^3} v_z^2 \tau \left(-\frac{\partial f_0}{\partial \varepsilon} \right) E.$$

Comparing this result with the definition of the conductivity, $j_z = \sigma E$, we see that

$$\begin{aligned} \sigma &= 2e^2 \int \frac{d^3p}{(2\pi)^3} v_z^2 v_z^2 \tau \left(-\frac{\partial f_0}{\partial \varepsilon} \right) \\ &= \frac{1}{3} e^2 \int d\varepsilon \nu(\varepsilon) v^2(\varepsilon) \tau(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right), \end{aligned}$$

where we used that for an isotropic Fermi surface $\langle \cos^2 \theta \rangle = 1/3$.

Noticing that the diffusion coefficient is equal to

$$D(\varepsilon) = \frac{1}{3} v^2(\varepsilon) \tau(\varepsilon),$$

we can re-write the conductivity in the Einstein's form

$$\sigma = e^2 \int d\varepsilon \nu(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right) D(\varepsilon).$$

Sometimes this result is re-written as

$$\sigma = \int d\varepsilon \sigma(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right),$$

where the *conductivity at fixed energy* is

$$\sigma(\varepsilon) \equiv e^2 \nu(\varepsilon) D(\varepsilon).$$

In a metal at $T = 0$,

$$-\frac{\partial f_0}{\partial \varepsilon} = \delta(\varepsilon - E_F)$$

so that

$$\sigma = \frac{1}{3} e^2 \nu_F v_F^2 \tau^F,$$

where index F indicates that the corresponding quantity is evaluated at the Fermi energy.

Our result can be cast into a Drude-like form. Indeed, recalling that

$$n = \frac{2}{3} \nu_F E_F,$$

we obtain

$$\sigma = \frac{e^2 n \tau^F}{m},$$

which is just the Drude formula.

A. General case of electron-impurity scattering

Probability w is related to the *differential scattering cross-section*, $dA/d\Omega$, of an impurity via

$$w_{\mathbf{p}' \rightarrow \mathbf{p}} = w_{\mathbf{p} \rightarrow \mathbf{p}'} = \frac{4\pi}{L^3} \frac{1}{\nu(\varepsilon)} \delta(\varepsilon - \varepsilon') v(\varepsilon) n_i dA/d\Omega,$$

where $\varepsilon \equiv p^2/2m$, $\nu(\varepsilon)$ is the density of states and n_i is the impurity concentration. The delta-function reflects the energy conservation. The block $n_i dA/d\Omega$ has a meaning of a differential scattering rate. The rest of the normalization factors is chosen for convenience. $dA/d\Omega$ can be calculated using either classical or quantum-mechanical scattering theory, depending on what is necessary in a particular situation. For example, if an impurity is a Coulomb center of charge Ze , then classical and quantum mechanics give identical results (see, e.g., Landau & Lifshits, *QM*)

$$dA/d\Omega = \left(\frac{Ze^2}{4\varepsilon} \right)^2 \frac{1}{\sin^4 \alpha/2},$$

where α is the angle between $\mathbf{p}$ and $\mathbf{p}'$. Notice a strong dependence of $dA/d\Omega$ both on ε and α –faster particles experience less scattering and scattering is predominantly by small angles. A neutral impurity can be modeled by a hard sphere of radius a . If $a \ll \lambda$, where λ is the wavelength of an incoming particle (see, e.g., Landau & Lifshits, *QM*),

$$[dA/d\Omega]_{QM} = 4 [dA/d\Omega]_{CM} = a^2,$$

where $[\dots]_{QM}$ and $[\dots]_{CM}$ stand for quantum-mechanical and classical results, respectively. In that case, $dA/d\Omega$ depends neither on ε nor on α ; the last condition means that scattering by any angle in the interval $(0, \pi)$ is equally probable.

The sum over momenta in the collision integral is transformed into an integral

$$\begin{aligned} I_{\text{coll}}[f] &= \sum_{\mathbf{p}'} w_{\mathbf{p} \rightarrow \mathbf{p}'} (f_{\mathbf{p}'} - f_{\mathbf{p}}) = n_i L^3 4\pi \int d^3 p' \frac{1}{L^3} \frac{1}{\nu(\varepsilon')} \delta(\varepsilon - \varepsilon') v(\varepsilon') dA/d\Omega (f_{\mathbf{p}'} - f_{\mathbf{p}}) \\ &= n_i \int d\varepsilon' \delta(\varepsilon - \varepsilon') v(\varepsilon') 4\pi \int \frac{d\phi' d\theta' \sin \theta'}{4\pi} dA/d\Omega (f_{\mathbf{p}'} - f_{\mathbf{p}}) \\ &= n_i v(\varepsilon) \int d\phi' d\theta' \sin \theta' dA/d\Omega (f_{\mathbf{p}'} - f_{\mathbf{p}}), \end{aligned} \quad (2.1)$$

where we have chosen a spherical coordinate system with $\mathbf{p}$ being the z-axis. $dA/d\Omega$ is a function of θ' but not ϕ' . θ' is the scattering angle.

Representing, as before, the distribution function as $f = f_0 + \mathbf{E} \cdot \mathbf{v} \chi$, the collision integral becomes

$$I_{\text{coll}}[f] = n_i \chi(\varepsilon) v \int d\phi' d\theta' \sin \theta' dA/d\Omega (\mathbf{E} \cdot \mathbf{v}' - \mathbf{E} \cdot \mathbf{v}),$$

where the magnitudes of $\mathbf{v}$ and $\mathbf{v}'$ are already the same. Denote the angle between $\mathbf{v}$ and $\mathbf{E}$ by θ_E . By the three-dimensional cosine theorem,

$$\cos(\mathbf{E}, \mathbf{v}') = \cos \theta_E \cos \theta' + \sin \theta_E \sin \theta' \cos(\phi' - \phi).$$

The integral over ϕ' nullifies the second term in $\cos(\mathbf{E}, \mathbf{v}')$ and we obtain

$$\begin{aligned} I_{\text{coll}}[f] &= n_i \chi(\varepsilon) v \int d\Omega' dA/d\Omega (\mathbf{E} \cdot \mathbf{v} \cos \theta' - \mathbf{E} \cdot \mathbf{v}) \\ &= -n_i \mathbf{E} \cdot \mathbf{v} \chi(\varepsilon) v \int d\Omega dA/d\Omega (1 - \cos \theta) \\ &= -n_i \mathbf{E} \cdot \mathbf{v} \chi(\varepsilon) v A_{\text{tr}}, \end{aligned}$$

where

$$A_{\text{tr}} \equiv \int d\Omega dA/d\Omega (1 - \cos \theta)$$

is the *transport cross-section* and in the last line we re-labeled $\theta' \rightarrow \theta$. The difference between the total cross-section $A_{\text{tot}} \equiv \int d\Omega dA/d\Omega$ and A_{tr} is in the weighing factor $1 - \cos \theta$. The meaning of this factor is obvious: it discriminates against small angle-scattering events which contributes less to the resistivity. For example, it is obvious then if scattering is strictly forward, i.e., $dA/d\Omega \propto \delta(\theta)$, conductivity must be infinite because all electrons

entering via one contact leave the other with probability 1. In that case, A_{tot} is finite but $A_{tr} = 0$. If scattering is predominantly by small angles,

$$A_{tr} = \int d\Omega dA/d\Omega (1 - \cos \theta) \approx \int d\Omega dA/d\Omega \frac{\theta^2}{2} \ll \int d\Omega dA/d\Omega = A_{tot}.$$

If $dA/d\Omega$ does not depend on θ (short-range impurity) then the $\cos \theta$ term is nullified when integrating over θ and

$$A_{tr} = A_{tot}.$$

The Boltzmann equation now takes the form

$$\begin{aligned} -e\mathbf{E} \cdot \mathbf{v} \frac{\partial f_0}{\partial \varepsilon} &= -n_i \mathbf{E} \cdot \mathbf{v} \chi(\varepsilon) v A_{tr} \rightarrow \\ \chi(\varepsilon) &= e \frac{\partial f_0}{\partial \varepsilon} \frac{1}{n_i v A_{tr}} \end{aligned}$$

Repeating the same steps as in the τ - approximation, we find

$$\begin{aligned} \sigma &= 2e^2 \int \frac{d^3p}{(2\pi)^3} v_z^2 \tau(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right) \\ &= \frac{1}{3} e^2 \int d\varepsilon \nu(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right) v^2(\varepsilon) \frac{1}{n_i A_{tr} v(\varepsilon)} = \\ &= \frac{1}{3} e^2 \int d\varepsilon \nu(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right) v^2(\varepsilon) \tau_{tr}(\varepsilon). \end{aligned} \tag{2.2}$$

where we introduced the *transport time* via

$$\frac{1}{\tau_{tr}} \equiv n_i A_{tr} v(\varepsilon).$$

Noticing that the diffusion coefficient

$$D(\varepsilon) = \frac{1}{3} v^2(\varepsilon) \tau_{tr}(\varepsilon),$$

we can re-write the conductivity in the Einstein's form

$$\sigma = e^2 \int d\varepsilon \nu(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right) D(\varepsilon).$$

Sometimes this result is re-written as

$$\sigma = \int d\varepsilon \sigma(\varepsilon) \left(-\frac{\partial f_0}{\partial \varepsilon} \right),$$

where the *conductivity at fixed energy* is

$$\sigma(\varepsilon) \equiv e^2 \nu(\varepsilon) D(\varepsilon).$$

In a metal at $T = 0$,

$$-\frac{\partial f_0}{\partial \varepsilon} = \delta(\varepsilon - E_F)$$

so that

$$\sigma = \frac{1}{3} e^2 \nu_F v_F^2 \tau_{\text{tr}}^F,$$

where index F indicates that the corresponding quantity is evaluated at the Fermi energy.

Again, the result can be cast into a Drude-like form:

$$\sigma = \frac{e^2 n \tau_{\text{tr}}^F}{m},$$

which is just the Drude formula. Our gain is that a phenomenological scattering time is now replaced by a certain microscopic time, τ_{tr}^F .

III. ELECTRON-PHONON INTERACTION

A. Estimate of the scattering time in the high-temperature regime

For high temperatures, $T \gg T_D$, the phonon wavelength is short $\lambda \simeq a_0$, therefore, oscillations of each atom proceed almost independently of the others. Ions arranged into a periodic lattice do not lead to damping of electron motion; thermal vibrations of the ions do. The scattering cross-section for an ion can be identified with rms thermal amplitude of the ions position. The rms amplitude is estimate simply from

$$\begin{aligned} M\omega_0^2 \langle r^2 \rangle_T &\simeq T \\ A &\simeq \langle r^2 \rangle_T \simeq T/M\omega_0^2. \\ \frac{1}{\tau} &= n_i v_F A \simeq \frac{1}{a_0^3} \frac{1}{ma_0} \frac{T}{M\omega_0^2}. \end{aligned}$$

The prefactor can be estimated further by recalling that the harmonic potential is obtained by expanding the electron term in deviations from the equilibrium position

$$\begin{aligned} U(r) - U(r_0) &= U(r - r_0 + r_0) - U(r_0) = \frac{1}{2} U''(r_0) (r - r_0)^2 \equiv \frac{1}{2} M\omega_0^2 (r - r_0)^2 \\ U''(r_0) &\simeq \frac{e^2}{a_0^3} \simeq M\omega_0^2 \end{aligned}$$

Now,

$$\frac{1}{\tau} \simeq \frac{1}{a_0^3} \frac{1}{ma_0} \frac{T}{e^2/a_0^3} \simeq \frac{T}{me^2 a_0} \simeq T.$$

Thus, we arrive at a conclusion that in metals the electron-phonon scattering rate is of order T , the prefactor being a number of order one. In semiconductors (and semimetals), where $k_F \ll a_0^{-1}$, same estimates lead to

$$\frac{1}{\tau} \simeq k_F a_0 T.$$

These estimates are in a good agreement with the experiment.

1. Time scales for e-ph scattering.

Electron moves through a units cell in $a_0/v_F \simeq 10^{-8} \text{ cm}/10^8 \text{ cm/s} \simeq 10^{-16} \text{ s}$. Period of ionic vibrations $\simeq 10^{-13} \text{ s}$. By the time an electrons move from one ion to another, ions are frozen. An electron scatters from static displacements of ions from equilibrium position

–“thermal disorder”. Thus scattering is almost elastic (“quasi-elastic”). Each scattering event $T \gg T_D$ consists of emitting or absorbing a short wavelength phonon $q \simeq a_0^{-1} \simeq k_F$. Thus the momentum changes significantly in every scattering event. On the other hand, the electron energy cannot change by more than $T_D \ll T$, by assumption. It takes many acts of emission/absorption to relax energy.

B. Deformation potential

If $v(\mathbf{r})$ is the potential energy of an electron-ion interaction, the total potential energy of an electron in a crystal is

$$V(\mathbf{r}) = \sum_{n: \text{ over all ions}} v(\mathbf{r} - \mathbf{r}_n),$$

where $\mathbf{r}_n = \mathbf{r}_n^{eq} + \mathbf{u}_n$, $\mathbf{r}_n^{eq}$ is the equilibrium position of the n^{th} ion and $\vec{u}_n$ is its displacement. For long-wavelength oscillations, $\sum_n \rightarrow a_0^{-3} \int d\mathbf{r}'$, where a_0^3 is the volume of a unit cell, and

$$V(\mathbf{r}) = \frac{1}{a_0^3} \int d^3r' v(\mathbf{r} - \mathbf{r}' - \mathbf{u}(\mathbf{r}')).$$

For small displacements,

$$\begin{aligned} V(\mathbf{r}) &= \frac{1}{a_0^3} \int d^3r' v(\mathbf{r} - \mathbf{r}' - \mathbf{u}(\mathbf{r}')) \\ &\approx \frac{1}{a_0^3} \int d^3r' \mathbf{u}(\mathbf{r}') \cdot \vec{\nabla}_{\mathbf{r}'} v(\mathbf{r} - \mathbf{r}') \\ &= -\frac{1}{a_0^3} \int d^3r' v(\mathbf{r} - \mathbf{r}') \vec{\nabla}_{\mathbf{r}'} \cdot \vec{u}(\mathbf{r}'). \end{aligned}$$

The electron-ion interaction is of the short-range due to screening. Model it with

$$v(\mathbf{r} - \mathbf{r}') = -\Xi a_0^3 \delta(\mathbf{r} - \mathbf{r}'),$$

where Ξ is the energy parameter known as the **deformation potential constant**. In this model,

$$W(\mathbf{r}) = \Xi \vec{\nabla} \cdot \mathbf{u}.$$

Physical meaning: $\vec{\nabla} \cdot \mathbf{u}$ is the relative change in volume due to deformation. The electron-phonon interaction is thus proportional to the relative change in volume. This form of the interaction is called **deformation potential**. Ξ is a phenomenological constant which is either calculated microscopically or is obtained from fitting of the scattering time to the

experimental data. An order-of-magnitude estimate for Ξ is obtained noticing that it is a typical energy of electron-ion interaction at distances between the two of order of the atomic constant. Hence,

$$\Xi \simeq e^2/a_0 \simeq \text{few } eV.$$

Indeed, experimental values of Ξ are in this range for all materials (metals, semiconductors, insulators). For metals, $\Xi \simeq E_F$. Another feature of the deformation potential is that it vanishes for uniform displacements: for $\mathbf{u} = \text{const}$, $W = 0$. This is quite natural as a uniform displacement means a translation of a crystal as a whole which cannot change an electron's energy. Once we know the interaction energy, we can calculate the transition probability for an electron $W_{\mathbf{k} \rightarrow \mathbf{k}'}$ using the quantum-mechanical perturbation theory. (Notice that $W \ll E_F$ for $|\vec{\nabla} \cdot \mathbf{u}| \ll 1$, i.e., for long-wavelength oscillations). However, quantum mechanics by itself is not sufficient to determine scattering times (and resistivities) because we need to average QM probabilities over thermal distributions of phonons and electrons. The way this averaging is to be done is prescribed by the Boltzmann equation.

C. Hamiltonian of the electron-phonon interaction in a second-quantized form

Classical kinetic energy of a deformed crystal

$$E_K = \frac{1}{2} \sum_n M \left(\frac{\partial \mathbf{u}_n}{\partial t} \right)^2.$$

For long-wavelength deformations,

$$E_K = \frac{1}{2} M n \int d^3 r \left(\frac{\partial \mathbf{u}(\mathbf{r})}{\partial t} \right)^2 = \frac{1}{2} \rho \int d^3 r \left(\frac{\partial \mathbf{u}(\mathbf{r})}{\partial t} \right)^2,$$

where n is the number density of ions. The momentum of the n^{th} ion equals to $M \partial_t \mathbf{u}_n$ and its coordinate is $\mathbf{u}_n$. Corresponding quantum-mechanical operators satisfy the usual p,x commutation relation

$$[\hat{u}_n^\alpha, M \partial_t \hat{u}_l^\beta] = i \delta_{nl} \delta_{\alpha\beta},$$

where $\alpha, \beta = x, y, z$. In the continuum limit,

$$[\hat{u}^\alpha(\mathbf{r}, t), M \partial_t \hat{u}^\beta(\mathbf{r}', t)] = i \delta_{\alpha\beta} \delta_{\mathbf{r}, \mathbf{r}'} = i \delta_{\alpha\beta} a_0^3 \delta(\mathbf{r} - \mathbf{r}') = i \delta_{\alpha\beta} a_0^3 \frac{1}{L^3} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')}, \quad (3.1)$$

where a_0^3 is the unit cell volume. Expand $\hat{u}$ in a Fourier sum

$$\hat{u}^\alpha(\mathbf{r}, t) = \frac{1}{\sqrt{L^3}} \sum_{\mathbf{k}} \hat{u}_{\mathbf{k}}^\alpha e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_k t)} + (\hat{u}_{\mathbf{k}}^\alpha)^\dagger e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega_k t)}.$$

The second term is added because $\hat{u}$ is an operator corresponding to a real quantity and is therefore Hermitian. ω_k is an eigenmode with wavenumber $\mathbf{k}$. The time derivative

$$\partial u^\alpha / \partial t = \frac{1}{\sqrt{L^3}} \sum_{\mathbf{k}} (-i\omega_k) \left\{ \hat{u}_{\mathbf{k}}^\alpha e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_k t)} - (\hat{u}_{\mathbf{k}}^\alpha)^\dagger e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega_k t)} \right\}$$

The equal-time commutator

$$\begin{aligned} [\hat{u}^\alpha(\mathbf{r}, t), M \partial_t \hat{u}^\beta(\mathbf{r}', t) / \partial t] &= \frac{M}{L^3} \sum_{\mathbf{k}, \mathbf{k}'} \omega_{k'} \left[\hat{u}_{\mathbf{k}}^\alpha e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_k t)} + \text{H.c.}, -i \hat{u}_{\mathbf{k}'}^\beta e^{i(\mathbf{k}' \cdot \mathbf{r} - \omega_{k'} t)} + \text{H.c.} \right] \\ &= \frac{M}{L^3} \sum_{\mathbf{k}, \mathbf{k}'} -i \omega_{k'} \left[\hat{u}_{\mathbf{k}}^\alpha, \hat{u}_{\mathbf{k}'}^\beta \right] e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_k t + \mathbf{k}' \cdot \mathbf{r}' - \omega_{k'} t)} \\ &\quad + i \omega_{k'} \left[\hat{u}_{\mathbf{k}}^\alpha, (\hat{u}_{\mathbf{k}'}^\beta)^\dagger \right] e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_k t - \mathbf{k}' \cdot \mathbf{r}' + \omega_{k'} t)} + \dots \end{aligned}$$

This commutator must be equal to (3.1). This is only possible if

$$\left[\hat{u}_{\mathbf{k}}^\alpha, \hat{u}_{\mathbf{k}'}^\beta \right] = \left[(\hat{u}_{\mathbf{k}}^\alpha)^\dagger, (\hat{u}_{\mathbf{k}'}^\beta)^\dagger \right] = 0$$

and

$$\left[\hat{u}_{\mathbf{k}}^\alpha, (\hat{u}_{\mathbf{k}'}^\beta)^\dagger \right] = \frac{a_0^3}{2M\omega_k} \delta_{\alpha\beta} \delta_{\mathbf{k},\mathbf{k}'} = \frac{1}{2\rho\omega_k} \delta_{\alpha\beta} \delta_{\mathbf{k},\mathbf{k}'},$$

where $\rho = M/a_0^3$. Now we can introduce bosonic operators

$$\hat{\mathbf{u}}_{\mathbf{k}} = \sqrt{\frac{1}{2\rho\omega_k}} \hat{n} b_{\mathbf{k}}$$

satisfying the canonical commutation relations

$$\begin{aligned} [b_{\mathbf{k}}, b_{\mathbf{k}'}^\dagger] &= \delta_{\mathbf{k},\mathbf{k}'} \\ [b_{\mathbf{k}}, b_{\mathbf{k}'}] &= [b_{\mathbf{k}}^\dagger, b_{\mathbf{k}'}^\dagger] = 0. \end{aligned}$$

Here $\hat{n}$ is unit vector in the direction of the displacement, that is *polarization*. In terms of these operators,

$$\hat{\mathbf{u}}(\mathbf{r}, t) = \frac{i\Xi}{\sqrt{L^3}} \sqrt{\frac{1}{2\rho\omega_k}} \sum_{\mathbf{k}} \hat{n} b_{\mathbf{k}} e^{i(\mathbf{k}\mathbf{r} - \omega_k t)} + \text{H.c.}$$

and

$$\vec{\nabla} \cdot \hat{\mathbf{u}} = i \sum_{\mathbf{k}} \sqrt{\frac{1}{2\rho\omega_k L^3}} \hat{n} \cdot \mathbf{k} b_{\mathbf{k}} e^{i(\mathbf{k}\mathbf{r} - \omega_k t)} + \text{H.c.}$$

It is now obvious that in the deformation potential model electrons interact only with longitudinal phonons $\mathbf{u} \parallel \mathbf{k}$. For those phonons, $\hat{n} \cdot \mathbf{k} = k$ and

$$\hat{W}(\mathbf{r}) = \Xi \vec{\nabla} \cdot \hat{\mathbf{u}} = i\Xi \sum_{\mathbf{k}} \sqrt{\frac{1}{2\rho\omega_k L^3}} k b_{\mathbf{k}} e^{i(\mathbf{k}\mathbf{r} - \omega_k t)} + \text{H.c.}$$

$\hat{W}$ is an operator of electron's energy in the presence of the elastic deformation of a crystal. A classical form of energy for a many-electron system in the presence of external perturbation is $E = \int d^3r n W$, where n is the number density, or in terms of Fourier transforms, $E = \int d^3q n(q) W(-q)$. A second-quantized form of the number density is $\hat{n}_{\mathbf{q}} = \sum_{\mathbf{p}} a_{\mathbf{p}+\mathbf{q}}^\dagger a_{\mathbf{p}}$. A corresponding second-quantized form is

$$H_{e-ph} = i\Xi \sum_{\mathbf{p}, \mathbf{k}} \sqrt{\frac{1}{2\rho\omega_k L^3}} k a_{\mathbf{p}+\mathbf{k}}^\dagger a_{\mathbf{p}} (b_{\mathbf{k}} + b_{\mathbf{k}}^\dagger).$$

D. Boltzmann equation for electron-phonon scattering

Now as we have the Hamiltonian for the electron-phonon interaction, we can start calculating matrix elements and transition probabilities. The only problem is that we need to decide which transition probabilities we need. The Boltzmann equation tells us precisely which matrix elements determine the evolution of the electron distribution function. Let's construct the collision integral for electron-phonon scattering.

First, look at processes of the type

$$\mathbf{p}', \mathbf{q} \Leftrightarrow \mathbf{p}$$

which corresponds either to absorption of a phonon with momentum $\mathbf{q}$ by the electron with momentum $\mathbf{p}'$, the latter changing to $\mathbf{p}$, or to emission of phonon $\mathbf{q}$ by the electron with momentum $\mathbf{p}$, the latter changing to $\mathbf{p}'$. The contribution of these two processes to the collision integral is

$$I_{\text{st}}^{(1)} = 2\pi \sum_{\mathbf{p}', \mathbf{q}} \delta(\varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}'} - \omega_{\mathbf{q}}) \left[\left| \langle \mathbf{p} | \hat{W} | \mathbf{p}', \mathbf{q} \rangle \right|^2 f_{\mathbf{p}'} (1 - f_{\mathbf{p}}) - \left| \langle \mathbf{p}', \mathbf{q} | \hat{W} | \mathbf{p} \rangle \right|^2 f_{\mathbf{p}} (1 - f_{\mathbf{p}'}) \right].$$

We are going to treat the e-ph interaction to the lowest order of the perturbation theory (Born approximation). In the Born approximation, the quantum-mechanical probabilities of time-reversed processes are the same. This is also true in our case with the exception that our transition probabilities depend on the occupation number of phonons and absorption/emission processes involve different combinations of these numbers. So the two transition probabilities are not equal yet.

In addition, there are also processes of the type

$$\mathbf{p}, \mathbf{q} \Leftrightarrow \mathbf{p}'.$$

A corresponding contribution to the collision integral is

$$I_{\text{st}}^{(2)} = 2\pi \sum_{\mathbf{p}', \mathbf{q}} \delta(\varepsilon_{\mathbf{p}'} + \omega_{\mathbf{q}} - \varepsilon_{\mathbf{p}}) \left[\left| \langle \mathbf{p}, \mathbf{q} | \hat{W} | \mathbf{p}' \rangle \right|^2 f_{\mathbf{p}'} (1 - f_{\mathbf{p}}) - \left| \langle \mathbf{p}' | \hat{W} | \mathbf{p}, \mathbf{q} \rangle \right|^2 f_{\mathbf{p}} (1 - f_{\mathbf{p}'}) \right].$$

The transition probabilities $\left| \langle \dots | \hat{W} | \dots \rangle \right|^2$ split into two factors: one is the quantum-mechanical probability of a transition between two electron states and the other one is the statistical probability that a certain number of phonon states is occupied. To see how it works, look at one of the transition amplitudes

$$\langle \mathbf{p} | \hat{W} | \mathbf{p}', \mathbf{q} \rangle = \langle \mathbf{p} | i\Xi \sum_{\mathbf{q}} \sqrt{\frac{1}{2\rho\omega_q L^3}} q b_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}} + \text{H.c.} | \mathbf{p}', \mathbf{q} \rangle.$$

This is an absorption process in which a state which has N_q phonons in a state with wavenumber $\mathbf{q}$ is lowered by one. The matrix element of operator $b_{\mathbf{q}}$ is (see Appendix for the proof of this result)

$$\langle \dots N_q - 1 | b_{\mathbf{q}} | \dots N_q \dots \rangle = \sqrt{N_q}.$$

The matrix element of operator $b_{\mathbf{q}}^\dagger$ is equal to zero since no phonons are created in this process. The integral over $\mathbf{r}$ yields the momentum conservation

$$\int d^3r e^{-i\mathbf{p}\cdot\mathbf{r}} e^{i\mathbf{q}\cdot\mathbf{r}} e^{i\mathbf{p}'\cdot\mathbf{r}} = L^3 \delta_{\mathbf{p}, \mathbf{p}'+\mathbf{q}}.$$

Finally,

$$\left| \langle \mathbf{p} | \hat{W} | \mathbf{p}', \mathbf{q} \rangle \right|^2 = L^3 \delta_{\mathbf{p}, \mathbf{p}'+\mathbf{q}} \frac{\Xi^2 q^2}{2\rho\omega_q} N_q$$

Absorption probabilities is different only in that the Bose occupation factor enters as $N_q + 1$. The Kronecker symbols eliminates summation over $\mathbf{q}$. Therefore,

$$I_{\text{st}}^{(1)} = 2\pi \int \frac{d^3p'}{(2\pi)^3} \delta(\varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}'} - \omega_q) \frac{\Xi^2 q^2}{2\rho\omega_q} [N_q f_{\mathbf{p}'} (1 - f_{\mathbf{p}}) - (N_q + 1) f_{\mathbf{p}} (1 - f_{\mathbf{p}'})] |_{\mathbf{q}=\mathbf{p}-\mathbf{p}'}.$$

Likewise,

$$I_{\text{st}}^{(2)} = 2\pi \int \frac{d^3p'}{(2\pi)^3} \delta(\varepsilon_{\mathbf{p}'} + \omega_q - \varepsilon_{\mathbf{p}}) \frac{\Xi^2 q^2}{2\rho\omega_q} [(N_q + 1) f_{\mathbf{p}'} (1 - f_{\mathbf{p}}) - N_q f_{\mathbf{p}} (1 - f_{\mathbf{p}'})] |_{\mathbf{q}=\mathbf{p}'-\mathbf{p}}.$$

The total collision integral is the sum of the two

$$I_{\text{st}} = I_{\text{st}}^{(1)} + I_{\text{st}}^{(2)}.$$

1. High temperatures

For $T \gg \Theta_D$, the collision integral is simplified considerably. Typical phonons have wavenumbers $\simeq q_D$ and frequencies $\simeq \omega_D \ll T$. This means that the Bose factors can be taken in the classical limit

$$\frac{1}{e^{\omega_q/T} - 1} \approx \frac{T}{\omega_q}$$

and

$$N_q + 1 \approx N_q.$$

Also, two electron energies differ typically by $T \gg \omega_q$. This means that one can neglect ω_q in the delta-functions. With these simplifications, we obtain

$$\begin{aligned} I_{\text{st}} &= 4\pi T \int \frac{d^3 p'}{(2\pi)^3} \delta(\varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}'}) \frac{\Xi^2 q^2}{2\rho\omega_q^2} [f_{\mathbf{p}'} - f_{\mathbf{p}}] |_{\mathbf{q}=\mathbf{p}-\mathbf{p}'} \\ &= T \int d\varepsilon' \nu(\varepsilon') \delta(\varepsilon - \varepsilon') \int d\Omega' \frac{\Xi^2 q^2}{2\rho\omega_q^2} [f_{\mathbf{p}'} - f_{\mathbf{p}}] |_{\mathbf{q}=\mathbf{p}-\mathbf{p}'} \\ &= T \nu(\varepsilon) \int d\Omega' \frac{\Xi^2 q^2}{2\rho\omega_q^2} [f_{\mathbf{p}'} - f_{\mathbf{p}}] |_{\mathbf{q}=\mathbf{p}-\mathbf{p}'} \end{aligned}$$

Notice that what we obtained is nothing more than the collision integral for elastic scattering—same as for the electron-impurity interaction, cf. Eq.(2.1) —with an effective differential cross-section

$$n_i v(\varepsilon) \frac{dA}{d\Omega} \rightarrow \nu(\varepsilon) T \frac{\Xi^2 q^2}{2\rho\omega_q^2}.$$

We see that indeed the “effective number of impurities” is proportional to T , and that phonons provide “thermal” static disorder of the lattice. For acoustic phonons,

$$\omega_q = sq$$

and the dependence on q drops out

$$T \frac{\Xi^2 q^2}{2\rho\omega_q^2} = T \frac{\Xi^2}{2\rho s^2}.$$

Independence of q means independence of the scattering angle. Thus, phonons are equivalent to short-range impurities whose cross-section is independent of both the electron’s energy and scattering angle. For such impurities transport and total cross-section coincide, so do the corresponding times. Without repeating the calculation we already did for the case of impurities, we seemly write the result for the conductivity

$$\sigma = \frac{e^2}{3} \int d\varepsilon \left(-\frac{\partial f_0}{\partial \varepsilon} \right) \nu(\varepsilon) v^2(\varepsilon) \tau_{\text{tr}}(\varepsilon),$$

where

$$\frac{1}{\tau_{\text{tr}}} = n_i v(\varepsilon) \int d\Omega \frac{dA}{d\Omega} (1 - \cos \theta) = T \nu(\varepsilon) \frac{\Xi^2}{2\rho s^2}.$$

Check units

$$\left[\frac{1}{\tau} \right] = E = E \frac{1}{EL^3} \frac{E^2}{M/L^3 (L/t)^2}.$$

For

$$\Theta_D \ll T \ll E_F,$$

the derivative of the Fermi function can be replaced by the delta-function. This leads to a usual Drude formula

$$\sigma = \frac{ne^2 \tau_F}{m},$$

where

$$\frac{1}{\tau_F} = \nu(E_F) \frac{\Xi^2}{2\rho s^2} T,$$

Recall that in a metal

$$\begin{aligned} \nu(E_F) &\simeq mk_F; k_F a_0 \simeq 1 \\ \Xi &\simeq Ms^2 \simeq 1/ma_0^2 \end{aligned}$$

Then

$$\frac{1}{\tau_F} \simeq T.$$

What if have a low-density metal—such as a degenerate semiconductor or a semimetal (bismuth or graphite). In such materials $k_F a_0 \ll 1$. Also, the band mass is usually smaller than the bare one $m^* \ll m$. Then the above estimate changes to

$$\frac{1}{\tau} \simeq k_F a_0 \frac{m^*}{m} T \ll T.$$

E. Low temperatures

We now turn to the case of low temperatures: $T \ll \Theta_D$. Technically, this is a more difficult case and a solution of the Boltzmann equation is not that straightforward as it is for higher temperatures. Instead of really solving the Boltzmann equation and finding the conductivity, we'll confine our treatment to finding the relaxation times and estimating, rather than calculating, the conductivity. The point is that e-ph scattering in this regime is inelastic, which means that the collision integral contains integrals over electron's energies

and the Boltzmann equation does not reduce to an algebraic equation for the correction to the distribution function. Some estimates" for $T \ll \Theta_D$, typical phonons have thermal energies: $\omega_q \simeq T$. Typical phonon momenta are $q \simeq T/s \ll \Theta_D/s \simeq a_0^{-1}s/s = a_0^{-1} \simeq k_F$ (in metals). Therefore, scattering is of the small-angle type. On the other hand, a typical electron's energy is within T from the Fermi energy: $|\varepsilon_p - E_F| \simeq T \simeq \omega_q$. Thus each e-ph scattering raises the electron's energy by $\simeq T$ which means that energy relaxation is fast.

Instead of setting up the problem for the conductivity, i.e., applying a weak electric field and finding a corresponding correction to the electron's distribution function, let's solve a different problem. Suppose that an electron of momentum $\mathbf{p}$ was injected into the system at $t = 0$. How long does it take for the system to relax back to an equilibrium state? Image that we produced a number of those electrons distributed homogeneously over the sample, then there are no spatial gradients of the distribution function. There are no external forces acting on the system either, so the Boltzmann equation takes the form

$$\frac{\partial f_{\mathbf{p}}}{\partial t} = I_{\text{st}}[f_p].$$

The rate at which $f_{\mathbf{p}}$ relaxes can be defined as

$$\frac{1}{\tau} \equiv -\frac{\delta}{\delta f_{\mathbf{p}}} \frac{\partial f_{\mathbf{p}}}{\partial t} = -\frac{\delta}{\delta f_{\mathbf{p}}} I_{\text{st}}[f_p].$$

As an example, consider the case of an exponential relaxation to the equilibrium state,

$$f_{\mathbf{p}} = f_{\mathbf{p}}^{(0)} + ae^{-t/\tau}.$$

Then,

$$\begin{aligned} \frac{\partial f_{\mathbf{p}}}{\partial t} &= -\frac{1}{\tau} ae^{-t/\tau} = -\frac{1}{\tau} (f_{\mathbf{p}} - f_{\mathbf{p}}^{(0)}) \\ -\frac{\delta}{\delta f_{\mathbf{p}}} \frac{\partial f_{\mathbf{p}}}{\partial t} &= \left(-\frac{1}{\tau}\right) \left(-\frac{\delta}{\delta f_{\mathbf{p}}}\right) (f_{\mathbf{p}} - f_{\mathbf{p}}^{(0)}) = \frac{1}{\tau}. \end{aligned}$$

The full form of the e-ph collision integral is

$$\begin{aligned} I_{\text{st}} &= 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}-\mathbf{q}} \delta(\varepsilon_p - \varepsilon_{p'} - \omega_q) [N_q f_{p'} (1 - f_p) - (N_q + 1) f_p (1 - f_{p'})] \\ &\quad 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}+\mathbf{q}} \delta(\varepsilon_p - \varepsilon_{p'} + \omega_q) [(N_q + 1) f_{p'} (1 - f_p) - N_q f_p (1 - f_{p'})], \end{aligned}$$

where

$$w_q \equiv \frac{\Xi^2 q^2}{2\rho\omega_q}$$

Taking the variational derivative with respect to f_p , we obtain

$$\begin{aligned} \frac{1}{\tau} = & 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}-\mathbf{q}} \delta(\varepsilon_p - \varepsilon_{p'} - \omega_q) \delta_{pp'} (f_p - N_q) \\ & 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}+\mathbf{q}} \delta(\varepsilon_p - \varepsilon_{p'} + \omega_q) \delta_{pp'} (f_p - N_q - 1) \\ & + 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}-\mathbf{q}} \delta(\varepsilon_p - \varepsilon_{p'} - \omega_q) [N_q + 1 - f_{p'}] \\ & + 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}+\mathbf{q}} \delta(\varepsilon_p - \varepsilon_{p'} + \omega_q) [f_{p'} + N_q] \end{aligned}$$

The first two terms are obtained by varying $f_{p'}$ and the last two by varying f_p . In the first two terms $\mathbf{p} = \mathbf{p}'$ and thus $q = 0$. For acoustic phonons, $w_{q=0} = 0$ and the first two terms drop out. Thus,

$$\begin{aligned} \frac{1}{\tau} = & 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}-\mathbf{q}} \delta(\varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}'} - \omega_q) [N_q + 1 - f_{p'}] \\ & + 2\pi L^3 \sum_{\mathbf{p}', \mathbf{q}} w_q \delta_{\mathbf{p}', \mathbf{p}+\mathbf{q}} \delta(\varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}'} + \omega_q) [f_{p'} + N_q]. \end{aligned}$$

In this equation, $f_{p'}$ is an equilibrium distribution function which depends only on the magnitude of $\mathbf{p}'$ but not its direction. Resolving the momentum conservation and changing from a sum over $\mathbf{q}$ to an integral, we find

$$\begin{aligned} \frac{1}{\tau} = & 2\pi \int \frac{d^3 q}{(2\pi)^3} w_q \delta(\varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}-\mathbf{q}} - \omega_q) [N_q + 1 - f_{\varepsilon_{\mathbf{p}}-\omega_q}] \\ & + 2\pi \int \frac{d^3 q}{(2\pi)^3} w_q \delta_{\mathbf{p}', \mathbf{p}+\mathbf{q}} \delta(\varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}+\mathbf{q}} + \omega_q) [f_{\varepsilon_{\mathbf{p}}+\omega_q} + N_q]. \end{aligned}$$

The arguments of the delta-functions are

$$\begin{aligned} \varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}-\mathbf{q}} - \omega_q &= v_p q \cos \theta - \frac{q^2}{2m} - sq; \\ \varepsilon_{\mathbf{p}} - \varepsilon_{\mathbf{p}+\mathbf{q}} + \omega_q &= -v_p q \cos \theta - \frac{q^2}{2m} + sq, \end{aligned}$$

where $v_p \equiv p/m$. The q^2 term is small. Dropping it, we see that the both delta-functions give the same result for $\cos \theta = s/v_p \ll 1$. Integrating over angle θ ,

$$\int_0^\pi d\theta \sin \theta \delta(v_p q \cos \theta - sq) = \int_{-1}^{+1} dx \frac{1}{v_p q} \delta(x - s/v_p) = \frac{1}{v_p q}.$$

Now,

$$\frac{1}{\tau} = \frac{1}{\pi} \frac{\Xi^2}{2v_p \rho} \int dq q \frac{q^2}{\omega_q} [2N_q + 1 - f_{\varepsilon_{\mathbf{p}}-\omega_q} + f_{\varepsilon_{\mathbf{p}}+\omega_q}].$$

The role of the factor containing the Bose and Fermi distribution functions is simply to limit the phonon momenta by the thermal value: $q_T \simeq T/s$. To see this, consider an electron at the Fermi surface, $\varepsilon_p = E_F$. Then,

$$\begin{aligned} 2N_q + 1 - f_{\varepsilon_{\mathbf{p}} - \omega_q} + f_{\varepsilon_{\mathbf{p}} + \omega_q} &= \frac{2}{e^{\omega_q/T} - 1} + 1 - \frac{1}{e^{-\omega_q/T} + 1} + \frac{1}{e^{\omega_q/T} + 1} \\ &= \frac{2}{e^{\omega_q/T} - 1} + \frac{e^{-\omega_q/T}}{e^{-\omega_q/T} + 1} + \frac{1}{e^{\omega_q/T} + 1} \\ &= \frac{2}{e^{\omega_q/T} - 1} + \frac{2}{e^{\omega_q/T} + 1}. \end{aligned}$$

Each of the terms falls off exponentially for $\omega_q = sq \gg T$, and thus the integration range is effectively limited to $q \lesssim T/s$. Final touches:

$$\begin{aligned} \frac{1}{\tau} &= \frac{1}{\pi} \frac{\Xi^2}{v_F \rho s} \int_0^\infty dq q^2 \left(\frac{1}{e^{\omega_q/T} - 1} + \frac{1}{e^{\omega_q/T} + 1} \right) \\ &= \frac{1}{\pi} \frac{\Xi^2}{v_F \rho s} \left(\frac{T}{s} \right)^3 \int_0^\infty dx x^2 \left(\frac{1}{e^x - 1} + \frac{1}{e^x + 1} \right) \\ &= \frac{7}{2\pi} \zeta(3) \frac{\Xi^2}{v_F \rho s} \left(\frac{T}{s} \right)^3, \end{aligned}$$

where we used that

$$\begin{aligned} \int_0^\infty dx x^2 \frac{1}{e^x + 1} &= \frac{3}{2} \zeta(3) \\ \int_0^\infty dx x^2 \frac{1}{e^x - 1} &= 2 \zeta(3). \end{aligned}$$

Here is our main result: the relaxation rate of electron-phonon scattering scales as T^3 for $T \ll \Theta_D$.

The rate we found has nothing to do with the conductivity. Remember, scattering is of the small-angle type and therefore the $1 - \cos \theta$ factor in the transport rate is small. Imagine that we insert this factor into the equation for the scattering rate. θ is an angle between vectors $\mathbf{p}$ and $\mathbf{p}'$: $\sin \theta/2 = q/2p$. As $q \ll p$, $\theta \approx q/p \ll 1$ and $1 - \cos \theta \approx \theta^2/2 = (1/2)(q/p)^2$. Instead of $1/\tau$, we get

$$\frac{1}{\tau} = \frac{1}{2\pi} \frac{\Xi^2}{v_F \rho s} \int_0^\infty dq q^2 \left(\frac{1}{e^{\omega_q/T} - 1} + \frac{1}{e^{\omega_q/T} + 1} \right) \frac{q^2}{p_F^2} \propto T^5$$

instead of T^3 . Hence, the resistivity also scales as T^5 . This is a famous T^5 -law or Bloch-Gruneisen law.

IV. ELECTRON-ELECTRON SCATTERING

In addition to scattering at impurities and phonons, electrons in conductors also scatter from each other. The rate of such scattering depends very strongly on whether we are dealing with a non-degenerate (as in semiconductors) or degenerate (as in metals) electron system.

A. Non-degenerate (Maxwell-Boltzmann) gas

If a system is non-degenerate, than an e-e collision proceeds very much the same as it does in vacuum. The reason is that the occupation numbers of all electron states in a non-degenerate gas are small ($\propto e^{-\varepsilon_p/T}$) and therefore electrons don't get into each other way when choosing the initial and final states for scattering. Suppose that the we have two electrons with velocities $\mathbf{v}_1$ and $\mathbf{v}_2$ in vacuum. To find the scattering cross-section, we choose the reference frame center of mass of the two-electron system is stationary. The differential cross-section is given by the Rutherford formula

$$\frac{dA}{d\Omega} = \left(\frac{e^2}{mv^2} \right)^2 \frac{1}{\sin^4 \theta/2},$$

where v is the relative velocity of two electrons in this system. In a dilute electron gas and at thermal equilibrium, two things change. a) a velocity is replaced by the thermal velocity and b) the singularity of the differential cross-section at small scattering angles is healed by *screening* –a combined effect of other electrons on a given one. The resulting transport scattering time is estimated as

$$\begin{aligned} \frac{1}{\tau_{tr}} &\simeq nv_T \left(\frac{e^2}{mv_T^2} \right)^2 \int_{\theta_c}^{\pi} d\theta \sin \theta (1 - \cos \theta) \frac{1}{\sin^4 \theta/2} \\ &\simeq nv_T \left(\frac{e^2}{mv_T^2} \right)^2 \int_{\theta_c}^{\simeq 1} \frac{d\theta}{\theta} = nv_T \left(\frac{e^2}{mv_T^2} \right)^2 \ln \frac{1}{\theta_c}, \end{aligned}$$

where θ_c is the “screening angle”. For scattering at angles smaller than θ_c , the Rutherford-like singularity in the cross-section is regularized by screening effects. To estimate θ_c , assume that screening leads to an effective cut-off of the Coulomb potential at distances larger than the screening radius, a_D . Then if a particle is impinging on a scattering center at an impact parameter, ρ , larger than a_D , the Coulomb potential is equal to zero. Now we need to relate

θ to ρ . For small-angle scattering, a change in the transverse momentum, δp_y , is related to the classical force by

$$\delta p_y = \int dt F_y$$

Typical values: $F_y \simeq e^2/\rho^2$ and $t \simeq \rho/v$. On the other hand, $\delta p_y = mv\theta$. Then

$$mv\theta \simeq \frac{\rho e^2}{v \rho^2}$$

$$\theta \simeq \frac{e^2}{\rho \varepsilon}.$$

Now

$$\theta_c \simeq \frac{e^2}{a_D \varepsilon} \simeq \frac{e^2}{a_D T}$$

Completing our estimate for τ_{tr} ,

$$\frac{1}{\tau_{tr}} \simeq nv_T \left(\frac{e^2}{mv_T^2} \right)^2 \ln \frac{a_D T}{e^2} \simeq nv_T \left(\frac{e^2}{T} \right)^2 \ln \frac{a_D T}{e^2}$$

It turns out that a_D also depends on temperature, but this only changes a weak-log-dependence of τ_{tr} . Neglecting the log,

$$\frac{1}{\tau_{tr}} \simeq nv_T \left(\frac{e^2}{T} \right)^2 \propto \frac{1}{T^{3/2}}.$$

As temperature decreases, $\frac{1}{\tau_{tr}}$ increases as $T^{-3/2}$. This increase continues only down to $T \simeq E_F$. For $T \ll E_F$, the energy dependence of the cross-section saturates as typical energies of electron are very close to the Fermi energy. However, Pauli principle starts to play a dominant role and gives a strong (T^2) dependence of $1/\tau$.

B. Degenerate (Fermi-Dirac) gas

Let's consider a $T=0$. All electrons occupy states within the Fermi sphere. Suppose we add one more electron at energy $\varepsilon > E_F$ (above the Fermi sphere), and ask how long will the relaxation of this electron take place. In what follows, we will not be interested in a particular form of the scattering cross-section. Suffices to say that as long as small-angle divergencies are cured by screening, the T -dependence does not depend on the form of the actual interaction. Let the scattering rate entering the Boltzmann equation, $w(q)$, where q is the momentum transfer, is known and behaves sufficiently well so that all integrals of w

over q we encounter in our analysis converge. Let's count electrons energies from the Fermi energy,

$$\epsilon \equiv \varepsilon - E_F.$$

At finite T , $|\epsilon| \simeq T$. Electron ϵ interacts with some electron which is initially below the Fermi energy ($\epsilon_1 < 0$). They scatter into the states ϵ' and ϵ'_1 , respectively. Because the empty states are available only above the Fermi energy, $\epsilon' > 0$ and $\epsilon'_1 > 0$. Energy and momentum conservation

$$\epsilon + \epsilon_1 = \epsilon' + \epsilon'_1.$$

$$\mathbf{p} + \mathbf{p}_1 = \mathbf{p}' + \mathbf{p}'_1.$$

How many unknowns do we have? Once the energies and momenta of the initial states are fixed, the energy and momenta of one of the electrons in the final state final is given uniquely by the conservation law. ϵ and $\mathbf{p}$ are fixed because we follow the evolution of a give electron's state. The transition probability is obtained by integrating the scattering rate over the states of two electrons: the “target” electron (ϵ_1 and $\mathbf{p}_1$) in the initial state and the other one in the final state. Instead on integrating over the momenta of one of the final electrons, let's say, $\mathbf{p}'$, we can equally well integrate over the momentum transfer

$$\mathbf{q} \equiv \mathbf{p} - \mathbf{p}' = \mathbf{p}'_1 - \mathbf{p}_1.$$

Also, we do not need to integrate over both momentum and energy of the same electron, as the two are related via dispersion $\epsilon = p^2/2m - E_F$. With all the above taken into account, the inverse lifetime of state $\mathbf{p}$, ϵ can be written as

$$\frac{1}{\tau} = 2\pi\nu_F \int d\epsilon_1 \int \frac{d\Omega'}{4\pi} \int \frac{d^3q}{(2\pi)^3} w(q) \delta(\epsilon_p + \epsilon_1 - \epsilon_{\mathbf{p}-\mathbf{q}} - \epsilon_{\mathbf{p}_1+\mathbf{q}}).$$

It is convenient to introduce an energy transfer, ω , via

$$\omega = \epsilon_p - \epsilon_{\mathbf{p}-\mathbf{q}} = \epsilon_{\mathbf{p}_1+\mathbf{q}} - \epsilon_1.$$

Then the delta-function can be re-written as

$$\frac{1}{\tau} = 2\pi\nu_F \int d\epsilon_1 \int d\omega \int \frac{d\Omega'}{4\pi} \int \frac{dq q^2}{(2\pi)^3} w(q) \int d\Omega_q \delta(\epsilon_p - \epsilon_{\mathbf{p}-\mathbf{q}} - \omega) \delta(\epsilon_{\mathbf{p}_1+\mathbf{q}} - \epsilon_1 - \omega).$$

The purpose of this trick is to make an integration over angles Ω' and Ω_q easier. Now expand the arguments of the delta-functions

$$\epsilon_p - \epsilon_{\mathbf{p}-\mathbf{q}} - \omega = v_p q \cos \theta_q - \omega \approx v_F q \cos \theta_q - \omega;$$

$$\epsilon_{\mathbf{p}_1+\mathbf{q}} - \epsilon_1 - \omega = v_{p_1} q \cos \theta' - \omega \approx v_F q \cos \theta' - \omega.$$

Integrals over $\cos \theta_q$ and $\cos \theta'$ give a factor of $(v_F q)^{-1}$ each. Continuing,

$$\begin{aligned} \frac{1}{\tau} &= \frac{1}{4\pi v_F^2} \nu_F \int d\epsilon_1 \int d\omega \int_0^\infty dq w(q) \\ &= C \int d\epsilon_1 \int d\omega, \end{aligned}$$

where a constant factor $C = \nu_F \int_0^\infty dq w(q) / 4\pi v_F^2$. Now it is time to notice that integrals over ϵ_1 and ω are constrained. Indeed, as $\omega = \epsilon_{\mathbf{p}_1+\mathbf{q}} - \epsilon_1$ and $\epsilon_{\mathbf{p}_1+\mathbf{q}} > 0$ (final state), whereas $\epsilon_1 < 0$ (initial state), $\omega > 0$. On the other hand, because $\omega = \epsilon - \epsilon_{\mathbf{p}-\mathbf{q}}$ and $\epsilon_{\mathbf{p}-\mathbf{q}} > 0$, we have $\omega < \epsilon$. Therefore, values of ω are limited to

$$0 < \omega < \epsilon.$$

In addition,

$$\epsilon_1 = \epsilon_{\mathbf{p}_1+\mathbf{q}} - \omega > -\omega$$

$$\epsilon_1 < 0.$$

Thus,

$$\begin{aligned} \frac{1}{\tau} &= C \int_0^\epsilon d\omega \int_{-\omega}^0 d\epsilon_1 \\ &= \frac{1}{2} C \epsilon^2. \end{aligned}$$

Thus the lifetime of an electron above the Fermi surface goes as the square of its energy measured from the Fermi energy. Because our result is even in ϵ , an empty state with energy less than E_F by ϵ (*hole*) is going to live the same time. In addition, at finite T , typical $|\epsilon| \simeq T$ and thus

$$\frac{1}{\tau} \propto \max \{ \epsilon^2, T^2 \}$$

(The exact formula is

$$\frac{1}{\tau} \propto \epsilon^2 + (\pi T)^2.$$

The coefficient of proportionality is readily estimated by noticing that in metals the Fermi energy and energy of the Coulomb interaction are of the same order

$$E_F \simeq e^2/a \simeq e^2/a_0,$$

where a is typical distance between electrons. Hence there are only two energy scales in the problem: T and E_F , each of which has units of $1/\tau$. As $1/\tau$ must be proportional to T^2 , the only combination is

$$\frac{1}{\tau} \simeq \frac{T^2}{E_F}.$$

Notice that at low enough temperatures, the e-e scattering rate is larger than the electron-phonon one (T^2 vs T^5).

C. Absence of the electron-electron and electron-phonon contributions to the resistivity in a Galilean-invariant system

Our previous discussions of relaxation times for electron-phonon and electron-electron interactions have a serious flaw: namely, in a Galilean-invariant system (no lattice and no impurities) the resistivity is zero regardless of whether these interactions are present or not. For a system with electron-electron interactions only, this is particularly obvious. To measure resistance, we apply an electric field to the system, therefore electrons are subject to an external force. In the Drude formulation of the transport, it is tacitly assumed that the electric force is balanced off by the “friction force”, so that the electron gas as a whole moves with a time-independent velocity. However, if electrons interact only with themselves, there is no “friction

” acting on the center-of-mass—by the Third Law, internal interactions in a system of particles do not affect the motion of the center-of-mass which moves under the action of the external forces only. In our case, this external force is the electric force which is not balanced by anything. Therefore, the center-of-mass moves with an acceleration and we can never achieve a steady-state situation when a small dc electric field causes a small dc current. For a coupled system of electrons and phonons, the argument is somewhat more subtle. The momenta of electron and phonon subsystems are not conserved separately, however, the total momentum is conserved. A static electric field acts directly only on electrons, which start to accelerate. Because of the electron-phonon interaction, electrons drag phonons along

(the name for this effect is “phonon drag”). As a result, we have a coherent, accelerated motion of both electrons and phonons. A coherent motion of phonons means the sound wave is generated. So, instead of a slow drift of electron subsystem, damped by scattering at phonons—this is what we need for finite resistivity in the Ohmic regime, we have an accelerated motion of electrons accompanied by a sound wave. Therefore, one can never achieve finite resistivity without including processes that relax momentum. These processes are scattering at impurities and Umklapp processes, in which only quasi-momentum, rather than the momentum is conserved.

1. *Does momentum-conserving scattering affect the conductivity?*

Let’s discuss a situation when we have both electron-electron and electron-impurity interactions, but lattice effects are ignored so that there is Umklapp scattering. The right-hand-side of the Boltzmann equation

$$\frac{df_{\mathbf{p}}}{dt} = I_{e-e} + I_{e-i}$$

contains two collision integrals: I_{e-e} and I_{e-i} , describing the electron-electron and electron-impurity interactions, correspondingly.

Theorem 3 *If a certain quantity—called additive collision invariant— is conserved in a scattering property, then the integral of this quantity with the corresponding collision integral is equal to zero.*

For example, electron-electron collisions conserve the number of particles, their total momentum and total energy. Therefore, unity (1), momentum $\mathbf{p}$, and energy $\varepsilon_{\mathbf{p}}$ are the additive collision invariants and

$$\int d^3p I_{e-e} = 0; \tag{4.1}$$

$$\int d^3p I_{e-e} \mathbf{p} = 0; \tag{4.2}$$

$$\int d^3p I_{e-e} \varepsilon_{\mathbf{p}} = 0. \tag{4.3}$$

Recall that the collision integral is a function of a single momentum $\mathbf{p}$, which the momentum of the state whose evolution we describe in the left-hand side of the expression. A formal

proof of this theorem can be found, *e.g.*, in Ref.[1]. Here I give a very short version of this proof which does not rely on particular properties of the collision integral. Suppose that we have only e-e interactions. Then the total momentum of the system

$$\mathbf{P} = \int \frac{d^3p}{(2\pi)^3} \mathbf{p} f(\mathbf{p}, \mathbf{r}, t)$$

is conserved, which means that

$$\frac{d\mathbf{P}}{dt} = \int \frac{d^3p}{(2\pi)^3} \mathbf{p} \frac{df(\mathbf{p}, \mathbf{r}, t)}{dt} = 0.$$

Multiplying both sides of the equation $df/dt = I_{e-e}$ by $\mathbf{p}$, and integrating over d^3p , we obtain property (4.2). Properties (4.1) and (4.3) are proven in a similar way.[?]

Now, let's add impurity scattering and turn on a static electric field. The Boltzmann equation takes the following form (I assume that the electric field is weak and use the standard, $1/\tau$ form of I_{e-i}

$$\frac{df}{dt} = -e\mathbf{E} \cdot \mathbf{v} \frac{\partial f_0}{\partial \varepsilon} = I_{e-e} + \frac{\bar{f} - f}{\tau_{tr}},$$

where τ_{tr} is the transport time for electron-impurity scattering, which I assume to be independent of electron's energy. By assumption, I_{e-e} has not changed its form, once we added impurities—and this is a strong assumption; however, any effect of a correlation between e-e and e-i interactions is beyond the Boltzmann equation. Thus, we still have property (4.2). The electric current is

$$\mathbf{j} = -2e \int \frac{d^3p}{(2\pi)^3} \mathbf{v} f = -\frac{2e}{m} \int \frac{d^3p}{(2\pi)^3} \mathbf{p} f,$$

where I used that in the absence of a lattice, $\mathbf{p} = m\mathbf{v}$. In our previous treatment of the conductivity, we first found a correction to f , and then substituted the result into the expression for the current. However, if—by assumption— τ_{tr} does not depend on the energy, we can find the current directly from the Boltzmann equation: by multiplying both sides by $-2e\mathbf{v}$ and integrating over d^3p . Let's check how it works first for $I_{e-e} = 0$:

$$-2e \int \frac{d^3p}{(2\pi)^3} \mathbf{v} \left(-e\mathbf{E} \cdot \mathbf{v} \frac{\partial f_0}{\partial \varepsilon} \right) = -2e \int \frac{d^3p}{(2\pi)^3} \mathbf{v} \frac{\bar{f} - f}{\tau_{tr}} = -\frac{\mathbf{j}}{\tau_{tr}}.$$

The integral of $\bar{f}$ vanishes because $\bar{f}$ is isotropic in the momentum space. Extracting the proportionality coefficient between $\mathbf{j}$ and $\mathbf{E}$, we arrive at the usual expression for the conductivity

$$\sigma = \frac{ne^2\tau_{tr}}{m}. \quad (4.4)$$

Now, put I_{e-e} back in and perform the same manipulation

$$\begin{aligned}
2e^2 \int \frac{d^3p}{(2\pi)^3} \mathbf{v} (\mathbf{E} \cdot \mathbf{v}) \frac{\partial f_0}{\partial \varepsilon} &= -2e \int \frac{d^3p}{(2\pi)^3} \mathbf{v} I_{e-e} - \frac{\mathbf{j}}{\tau_{tr}} \\
&= -\frac{2e}{m} \underbrace{\int \frac{d^3p}{(2\pi)^3} \mathbf{p} I_{e-e}}_{=0 \text{ by (4.2)}} - \frac{\mathbf{j}}{\tau_{tr}} \\
&= -\frac{\mathbf{j}}{\tau_{tr}},
\end{aligned}$$

and the conductivity is again given by Eq. (4.4), *regardless of whether we have e-e interactions or not*. Therefore, at the level of Boltzmann equation, the electron-electron interaction does not affect the conductivity. Notice that we did not use any particular properties of I_{e-e} , except for it conserves the momentum. Therefore, our proof is valid for any momentum-conserving process.

What if τ_{tr} depends on energy? Than our trick of finding the current directly from the Boltzmann equation would not work because we will end up with the integral $\int d^3p f \mathbf{v} / \tau_{tr}(p)$ which is not equal to the current. I will refer you to the book by Gantmakher and Levinson [2], where this situation is studied in detail. In short, there is a contribution to the resistivity from e-e scattering which works as follows. There is a characteristic temperature at which the times of electron-electron and electron-impurity scatterings match

$$\tau_{e-e}(T^*) \sim \tau_{tr}.$$

For example, for $\tau_{e-e}(T) \sim E_F/T^2$,

$$T^* \sim \sqrt{E_F/\tau_{tr}}$$

and, as $E_F \tau_{tr} \gg 1, 1/\tau_{tr} \ll T^* \ll E_F$. Below T^* , the conductivity is given by Eq. (2.2), which we can re-write as

$$\sigma_{T \ll T^*} = \frac{ne^2}{m} \langle \tau_{tr} \rangle, \quad (4.5)$$

where

$$\langle \tau_{tr} \rangle = \int d\varepsilon \left(-\frac{\partial f_0}{\partial \varepsilon} \right) \nu(\varepsilon) v^2(\varepsilon) \tau_{tr}(\varepsilon) / \nu_F v_F^2. \quad (4.6)$$

In other words, the correct τ to be put in the Drude formula, is the one averaged over energies as specified by Eq. (4.6). For $T \gg T^*$, the proper procedure is to average $1/\tau_{tr}$

$$\left\langle \frac{1}{\tau_{tr}} \right\rangle = \int d\varepsilon \left(-\frac{\partial f_0}{\partial \varepsilon} \right) \nu(\varepsilon) v^2(\varepsilon) \frac{1}{\tau_{tr}(\varepsilon)} / \nu_F v_F^2$$

so that the conductivity is now given by

$$\sigma_{T \gg T^*} = \frac{ne^2}{m} \left\langle \frac{1}{\tau_{\text{tr}}} \right\rangle^{-1}. \quad (4.7)$$

Given that still $T \ll E_F$, both conductivities in Eqs. (4.5) and (4.7) do not depend on T , up to terms $\mathcal{O}(T^2/E_F^2)$. However, these two conductivities are *different*. For $T \sim T^*$, a crossover between two conductivities occur. Thus, the effect of electron-electron interactions is to interpolate between two temperature-independent limiting forms of σ . If $\tau_{\text{tr}}(\varepsilon)$ depends on ε as a power-law, the two limiting forms differ by a numerical prefactor which means that e-e interactions can account of 2-3 .. fold variation of the conductivity with temperature.

Another special case is a metal or semiconductor with two (or more) types of carriers, with different e/m ratio. In this case, total current is not proportional to total momentum. Indeed,

$$\mathbf{j} = -2 \left[\frac{e_1}{m_1} \int \frac{d^3p}{(2\pi)^3} \mathbf{p} f_1 + \frac{e_2}{m_2} \int \frac{d^3p}{(2\pi)^3} \mathbf{p} f_2 \right];$$

$$\mathbf{P} = \int \frac{d^3p}{(2\pi)^3} \mathbf{p} (f_1 + f_2)$$

and $\mathbf{j}$ is *not* proportional to $\mathbf{P}$. In this case, the electron-electron interaction provides *friction* between two (or more) components of electron gas and does show up in the final expression for the conductivity. Even more striking example, is a compensated metal (semi-metal, such as Bi or graphite), in which the number of electrons is equal to the number of holes. In this case, one can have the resistivity from electron-electron interactions only (without Umklapps!). Indeed, the resistivity of Bi shows a well-pronounced T^2 part although the number of carriers in this semimetal is extremely small (10^{-5} per unit cell) and the Fermi surface occupies only a tiny fraction of the Brillouin zone.

Now, let's turn to the electron-phonon interaction. In fact, we did obtain the finite conductivity in this case by actually solving the Boltzmann equation rather than calculating just the scattering time and then substituting it into the Drude formula (which, for electron-electron interaction case would lead to a wrong result). Why did not have problems with the momentum conservation? If you look at our solution, what we assumed was that phonons are described by the *equilibrium* distribution function, n_0 . There is an implicit assumption here, that while we are driving electrons with the electric field, there is some relaxational

mechanism by which phonons maintain their equilibrium. If this is so, our result for the conductivity is correct. However, what is this hidden relaxational mechanism? If our model consists only of electrons and phonons, and Umklapps are forbidden, the momentum is conserved in all scattering processes: e-e, e-ph, and ph-ph, so that net force acting a coupled system of electrons and phonons will drive it out of the equilibrium. Thus we need to do away with the assumption about equilibrium in the phonon system, and describe them with their own Boltzmann equation, so that the full description is provided by a *system* of Boltzmann equations

$$\begin{aligned}\frac{df}{dt} &= I_{\text{e-ph}} + I_{\text{e-i}}; \\ \frac{dn}{dt} &= I_{\text{ph-e}} + I_{\text{ph-ph}} + I_{\text{ph-i}},\end{aligned}$$

where I discarded the e-e collisions, as we have already learned how to deal with them, but added impurities, as we will need them to maintain the current. Now, each of the e-ph and ph-e collision integrals does not conserve the momentum, but their *sum* does

$$\int d^3p \mathbf{p} I_{\text{e-ph}} + \int d^3k \mathbf{k} I_{\text{ph-e}} = 0. \quad (4.8)$$

Also, the ph-ph integral conserves the phonon momentum on its own

$$\int d^3k \mathbf{k} I_{\text{ph-ph}} = 0. \quad (4.9)$$

Again, turn on a static electric field and disregard all inhomogeneities. As the electric field couples only to electrons, the left-hand side of the Boltzmann equation for phonons is equal to zero: dn/dt . And again, let's choose the simplest forms of impurity integrals—for phonons it's going to be of the same form as for electrons but with a different scattering time. Then

$$\begin{aligned}-e\mathbf{E} \cdot \mathbf{v} \frac{\partial f_0}{\partial \varepsilon} &= I_{\text{e-ph}} + \frac{\bar{f} - f}{\tau_{\text{tr}}}; \\ 0 &= I_{\text{ph-e}} + I_{\text{ph-ph}} + \frac{\bar{n} - n}{\tau_{\text{tr}}^{\text{ph}}}.\end{aligned}$$

Multiplying the first equation by $-2e\mathbf{p}/m$ and the second one by $-2e\mathbf{k}/m$, taking into account that $\bar{f}$ and $\bar{n}$ is isotropic in the momentum space, adding up the two equations, and using (4.8) and (4.9), we obtain

$$2e^2 \int \frac{d^3p}{(2\pi)^3} \mathbf{v} (\mathbf{E} \cdot \mathbf{v}) \frac{\partial f_0}{\partial \varepsilon} = -\frac{\mathbf{j}}{\tau_{\text{tr}}} - \frac{\mathbf{J}}{\tau_{\text{tr}}^{\text{ph}}}, \quad (4.10)$$

where

$$\mathbf{J} = -\frac{2e}{m} \int d^3k \mathbf{k} n$$

looks like a part of the electric current carried by phonons. Now we need to compare τ_{tr} with $\tau_{\text{tr}}^{\text{ph}}$. If we recall, why the sky is blue it's because the scattering cross-section of light on microscopic obstacles in the atmosphere (water vapor, dust, etc.) is large for smaller wavelengths, hence the scattered spectrum is shifted to the blue side. Raleigh showed that the dependence of the cross-section on the wavelength is very strong: $A \propto \lambda^{-4}$. This mechanism is actually not specific for light but occurs any time when we are dealing with scattering of a wave on an object of size small compared to λ . For phonons at low T , $\lambda \sim s/T$, hence $1/\tau_{\text{tr}}^{\text{ph}} \propto T^4 \ll 1/\tau_{\text{tr}}$. Discarding then the second term in the right-hand side of Eq. (4.10), we arrive at the same relation between the current and the electric field as in the absence of both e-e and e-ph interactions. Notice though, that keeping this term would result in a T^4 -dependence of σ , which, I remind, comes neither from e-ph nor ph-ph scattering but momentum-relaxing ph-i processes.

A comment is in order here. In the discussion presented above, we tacitly assumed that electrons and phonons occupy the same space. This is not so in semiconductor heterostructures, when a plane or wire of electron gas is embedded into a three-dimensional crystal. In this case, electrons work as emitters of phonons which interact and eventually relaxed by the whole continuum of 3D phonon states, playing the role of a thermostat. In this case, the equilibrium in the phonon system is maintained due to coupling to a thermostat and finite resistance due to electron-phonon scattering is achieved without Umklapps or impurities.

2. Umklapp processes

Notice that in our treatment we neglected the lattice effects, i.e., we used the plane waves for electron states. In fact, we should have used Bloch states

$$\psi_{\mathbf{k}}(\mathbf{r}) = u_{\mathbf{k}}(r) e^{i\mathbf{k}\cdot\mathbf{r}},$$

where $u_{\mathbf{k}+\vec{b}} = u_{\mathbf{k}}$ and $\vec{b}$ is the reciprocal lattice vector. Since instead of the real momenta electrons have quasi-momenta, the momentum conservation is replaced by the quasi-momentum conservation

$$\mathbf{k}' = \mathbf{k} + \mathbf{q} + \vec{b}, \tag{4.11}$$

where $\vec{b}$ is the reciprocal lattice vector. Processes in which $\vec{b} = 0$ are called *normal* processes and those in which $\vec{b} \neq 0$ are called *Umklapp* processes. In a good metal, $k \approx k' \approx k_F$ and $k_F \simeq b$. For $T \gg \Theta_D$, $q \simeq q_D \simeq b$. This means that all three quasi-momenta in the conservation law are of order of b and there is no problem in finding $\mathbf{k}, \mathbf{k}'$, and $\mathbf{q}$ satisfying the (4.11) with $b \neq 0$. Thus normal and Umklapp processes are roughly equally probable. Taking Umklapp processes into account will change a prefactor in our result $1/\tau = aT$ but not the T -dependence. On the other hand, in a low density material $k_F \ll b$. Also, for $T \ll \Theta_D$, typical (acoustic) phonons have wavenumbers $q \simeq T/s \ll b$. This means that (4.11) cannot be satisfied with electrons near the Fermi surface and thermal phonons: Umklapp processes require that at least one of the momenta is $\simeq b$. Since the number of electrons and phonons with such momenta is exponentially small, the probability of Umklapp processes are also exponentially small: $(1/\tau)_{Umklapp} \propto \exp(-1/T)$. The exponential rise of the conductivity at low T was indeed observed in ultra-clean alkaline metals with closed Fermi surface. If the Fermi surface is open, *i.e.*, it intersects the Brillouin zone boundaries, Umklapp scattering is not distinguishable from normal one and resistivity is finite.

V. APPENDIX: SOME RESULTS OF THE SECOND QUANTIZATION FORMALISM

Let's prove that

$$\langle N-1 | b | N \rangle = \sqrt{N}; \quad (5.1)$$

$$\langle N+1 | b^\dagger | N \rangle = \sqrt{N+1}, \quad (5.2)$$

where I use a short-hand notation $N_q \equiv N, b \equiv \hat{b}_q$ and where it is understood that b and $b^\dagger$ act only on the occupation number of state q . Also, states which are not affected by b and $b^\dagger$ are skipped in the notation of the state-vector. Suppose that b are boson operators satisfying

$$bb^\dagger - b^\dagger b = 1.$$

Sandwich the commutation relation between states $|N\rangle$. Using the rule of matrix multiplication,

$$\langle N | b | M \rangle \langle M | b^\dagger | N \rangle - \langle N | b^\dagger | M \rangle \langle M | b | N \rangle = 1,$$

where the sum over M is implied.

By the definition of a Hermitian conjugate,

$$\langle M|b^\dagger|N\rangle = \int dq \psi_M^* (b^*)^T \psi_N = \int dq \psi_N b^* \psi_M^* = \left(\int dq \psi_N^* b \psi_M \right)^* = (\langle N|b|M\rangle)^*$$

and the commutation relation reduces to

$$|\langle N|b|M\rangle|^2 - |\langle M|b|N\rangle|^2 = 1.$$

Since b and change the occupation numbers by -1 , the only non-zero matrix elements contributing to the sum are

$$|\langle N|b|N+1\rangle|^2 - |\langle N-1|b|N\rangle|^2 = 1$$

Without a loss in generality, we can assume that the matrix elements are real. Then,

$$\langle N|b|N+1\rangle^2 - \langle N-1|b|N\rangle^2 = 1$$

Start with $N = 0$. As $b|0\rangle = 0$,

$$\langle 0|b|1\rangle^2 = 1.$$

For $N = 1$,

$$\langle 1|b|2\rangle^2 = \langle 0|b|1\rangle^2 + 1 = 2.$$

Repeating this process N times, we see that

$$\langle N|b|N+1\rangle^2 = N$$

and

$$\langle N|b|N+1\rangle = \sqrt{N},$$

which is equivalent to statements (5.1,5.1) .

[1] E. M. Lifshits and L. P. Pitaevskii, *Physical Kinetics*

[2] V. F. Gantmakher and Y. B. Levinson, *Carrier scattering in metals and semiconductors*, (Elsevier, North Holland, 1986).